

# GitHub & Engineering Portfolio

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## Public Repositories & Live Products

### DoWhiz - Digital Assistant Platform (Rust/TypeScript)

[github.com/KnoWhiz/DoWhiz](https://github.com/KnoWhiz/DoWhiz)

- Designed **multi-tenant isolation** with per-user workspace scoping and sandboxed CLI tools for secure concurrent user sessions
- Built **Discord CLI integration** for college student group work scenarios; completed full feature requirements and conducted real user testing with university students
- Developed **Google Workspace integration** for autonomous Drive management, document commenting, and Calendar/Meet scheduling
- Implemented **Notion webhook + MCP server** for page editing, task boards, and cross-channel follow-ups
- Migrated AI backend to **Claude Agent SDK** with structured thinking display; designed multi-turn session memory with file-based locking
- Managed **GitHub Actions CI/CD**; optimized **Azure ACI** container orchestration; debugged production VMs with CosmosDB query analysis

### DeepTutor - AI Learning Platform (Live Product)

[www.deep-tutor.com](http://www.deep-tutor.com)

- Optimized **agentic flow architecture** for intelligent tutoring, improving response quality and context handling
- Redesigned **file upload system** with chunked processing and progress tracking for large document support
- Architected **document sync infrastructure** achieving 50-100x speedup with incremental indexing; implemented Azure parallel upload reducing sync time from 190s to 15-20s for 1000-file libraries

## Personal Projects

[github.com/guihunwansui](https://github.com/guihunwansui)

- **CV** - Portfolio and resume materials (Jupyter Notebook)
- **SimpleNet** - Neural network implementation (Python)
- **DST-SmartBaseManager** - Game server management tool (Lua)

## Technical Contributions Summary

| Area         | Technologies & Contributions                                   |
|--------------|----------------------------------------------------------------|
| Backend      | Rust (production services), Python, TypeScript, RESTful APIs   |
| Cloud/Infra  | Azure (VM, ACI, Blob, CosmosDB), GitHub Actions CI/CD, Docker  |
| AI/ML        | Claude Agent SDK, agentic workflow design, RAG architecture    |
| Integrations | Google Workspace API, Notion API, Discord API, webhook systems |
| Database     | MongoDB, PostgreSQL, CosmosDB, vector search (Pinecone)        |